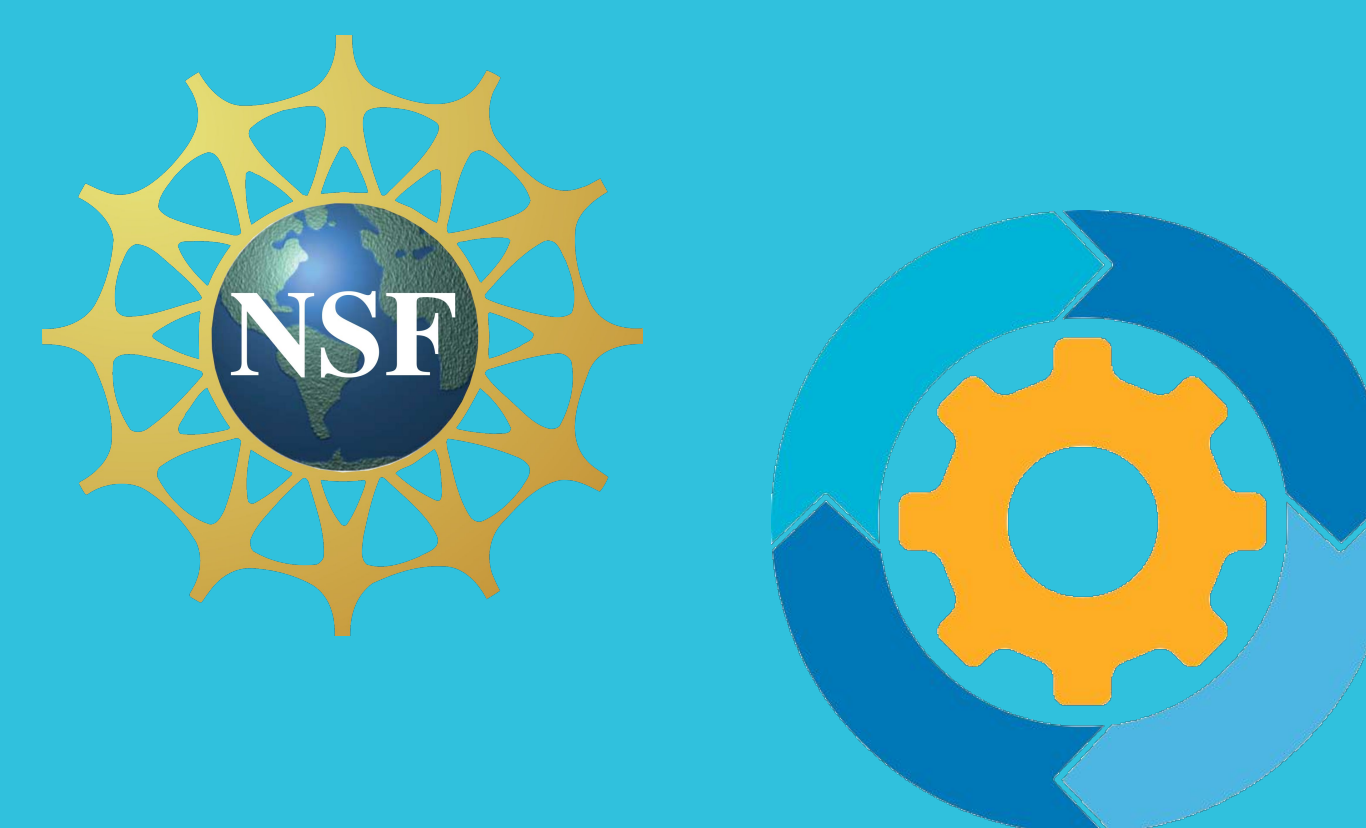


Experiences with Broadening CS Participation in Schools that Serve Indigenous Students



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Abstract

Indigenous-serving schools often **struggle in implementing culturally relevant computer science (CS)** in their curriculum. The project, **Let's Talk Code (LTC)** attempts to remedy this by providing professional development **workshops for indigenous-serving teachers** with the goal of increasing the confidence and competence in teaching culturally relevant computing and coding lessons into their curriculum.

Objective

Design professional development **workshops** for **indigenous-serving teachers** to learn effective techniques for relating **coding to culture**.

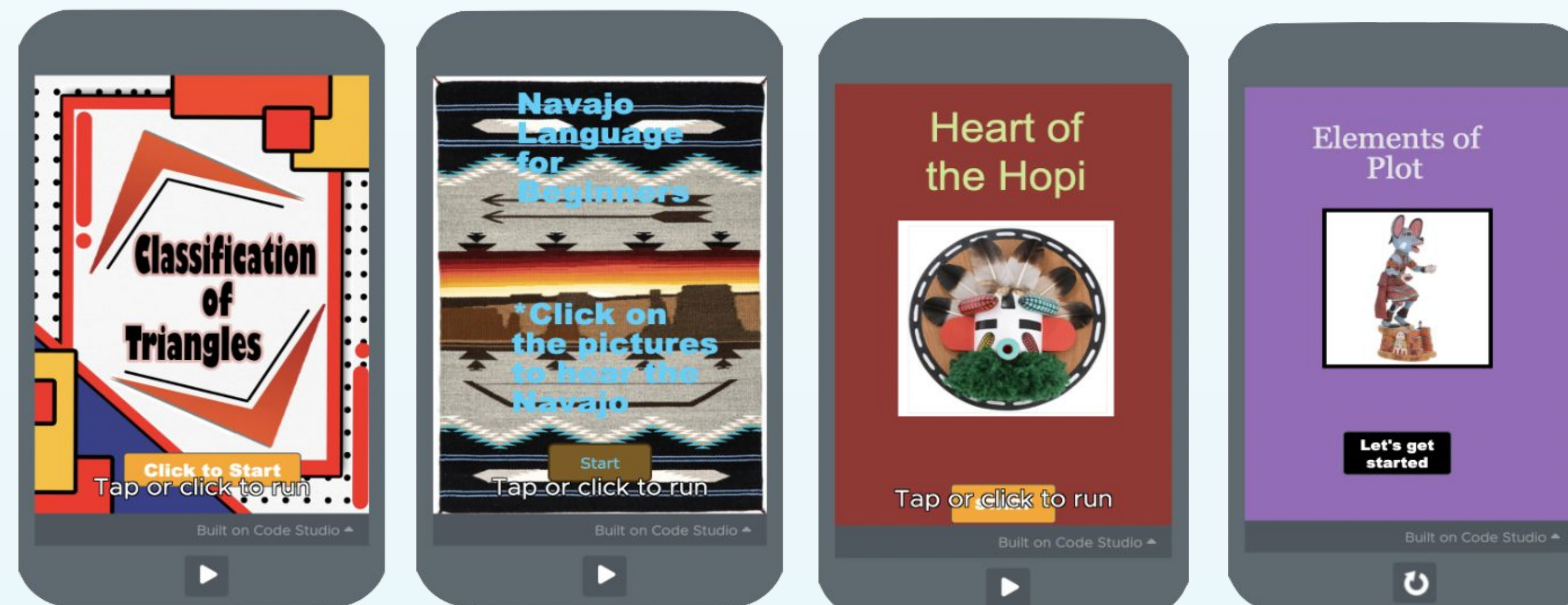


Background

The implementation of culturally relevant computer science curriculum has been a difficult task for schools that serve indigenous students. This issue could be caused by a variety of different reasons from the past to the present. The goal of Let's Talk Code (LTC) is to help educators in indigenous-serving schools to gain the confidence and the skills to teach culturally relevant programming lessons into their curriculum by providing professional development workshops.

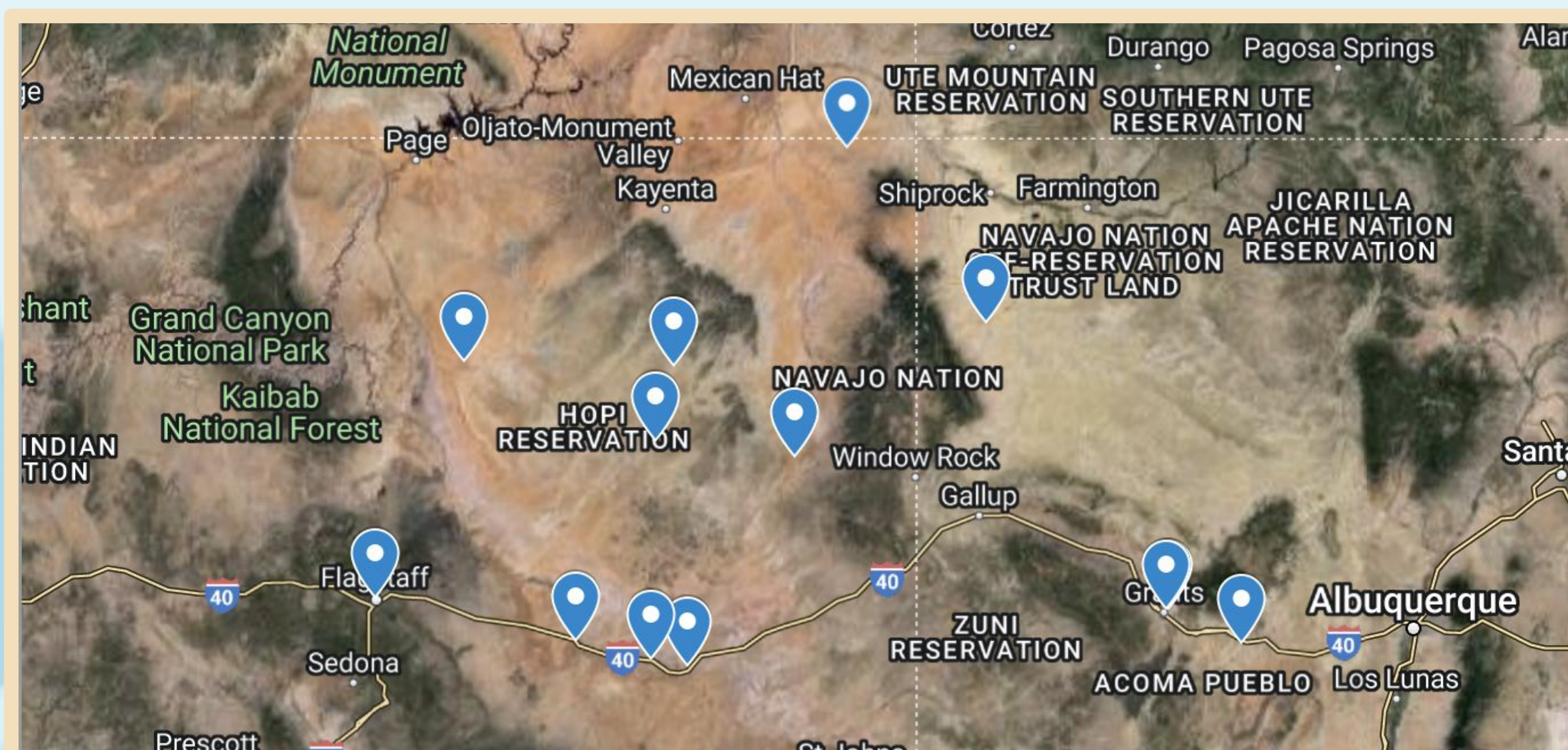
Program Description

- A Research-Practice Partnership (RPP) model collaborating with indigenous-serving schools in AZ and NM
- Teachers participate in 40 hours of:
 - **Coding basics**
 - How to implement **CS** with **culturally relevant pedagogy**
- Teachers implement a CS lesson plan into their curriculum
- Survey being completed by teachers and students before and after implementation



Collaboration

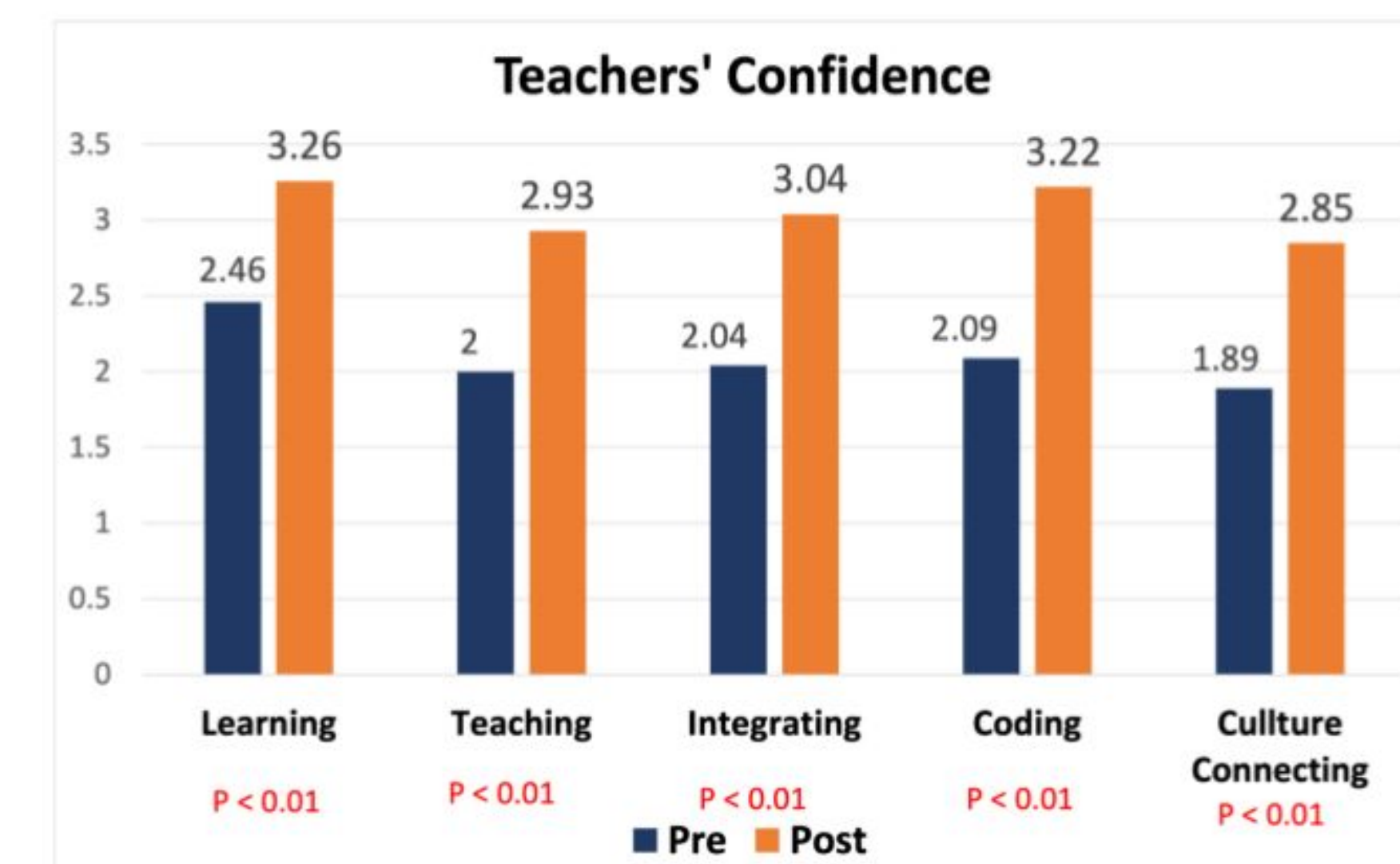
This project and paper followed the **Research-Practice Partnership** in order to foster a positive and collaborative relationship between CS experts and indigenous serving schools. This approach helped to create culturally responsive computer science lesson plans and implementations.



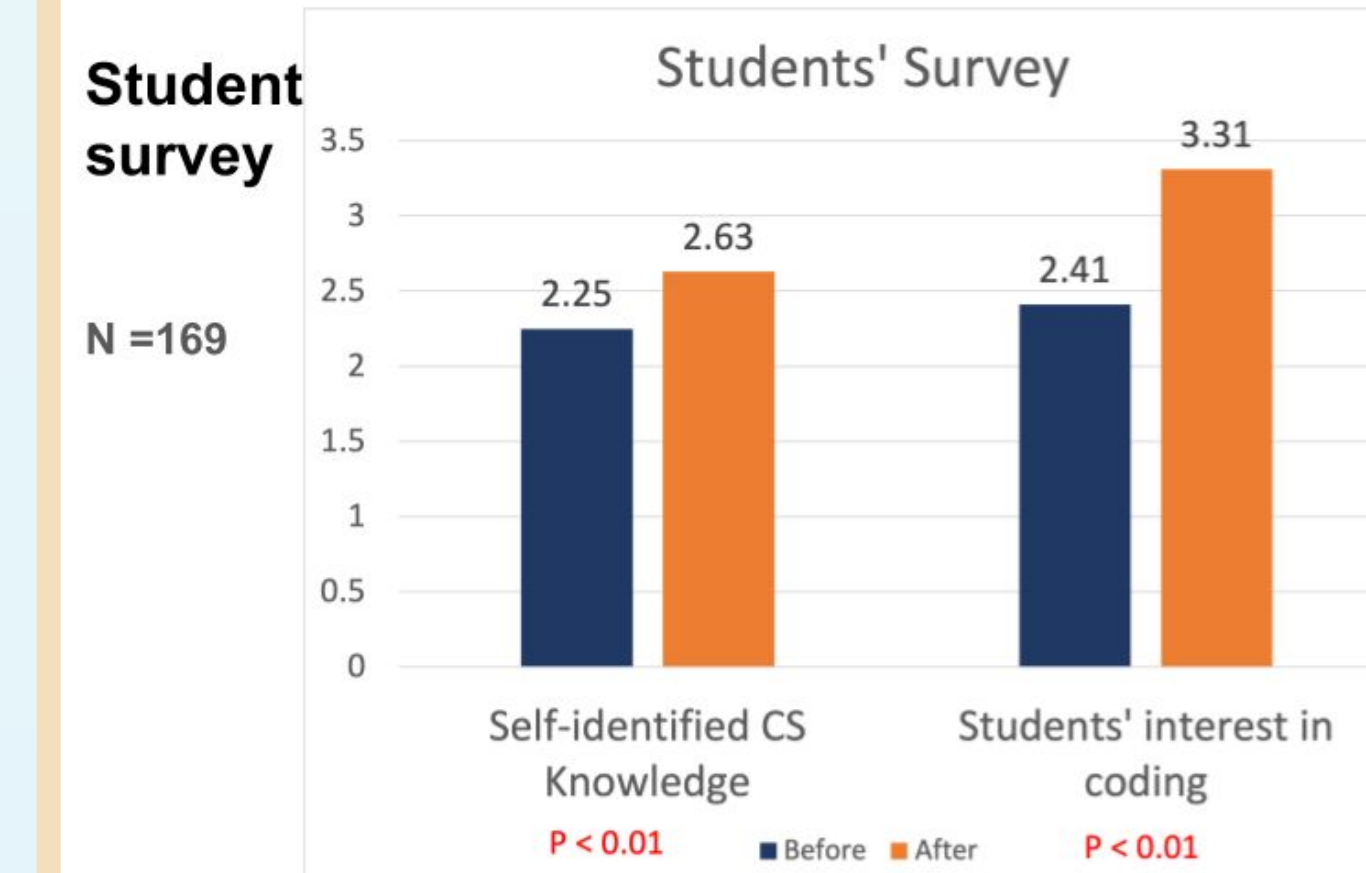
Preliminary Results

The evaluation of data collected through surveys reveals that there is a slight increase in confidence post professional development in regard to creating connections between coding and relating it to culturally relevant material. Students had also shown an increase of interest and confidence after teacher implementations.

Teachers' Confidence Survey N =46



Student survey



View Lesson Plans



Future Plans

- Refining **data collection** from teachers and students
- Developing a more empowering professional development experience to **increase teacher confidence** in creating culturally relevant computer lesson plans.